

Basics of Embedded Control System

What is Embedded System

- An **Embedded System** is a system that has software embedded into computer-hardware, which makes a system dedicated for a variety of application or specific part of an application or product or part of a larger system.
- It can be a small independent system or a large combinational system.
- It is a microcontroller-based control system used to perform a specific task of operation.

Major components of Embedded Control systems

- **Three major components:**

- **Hardware:**

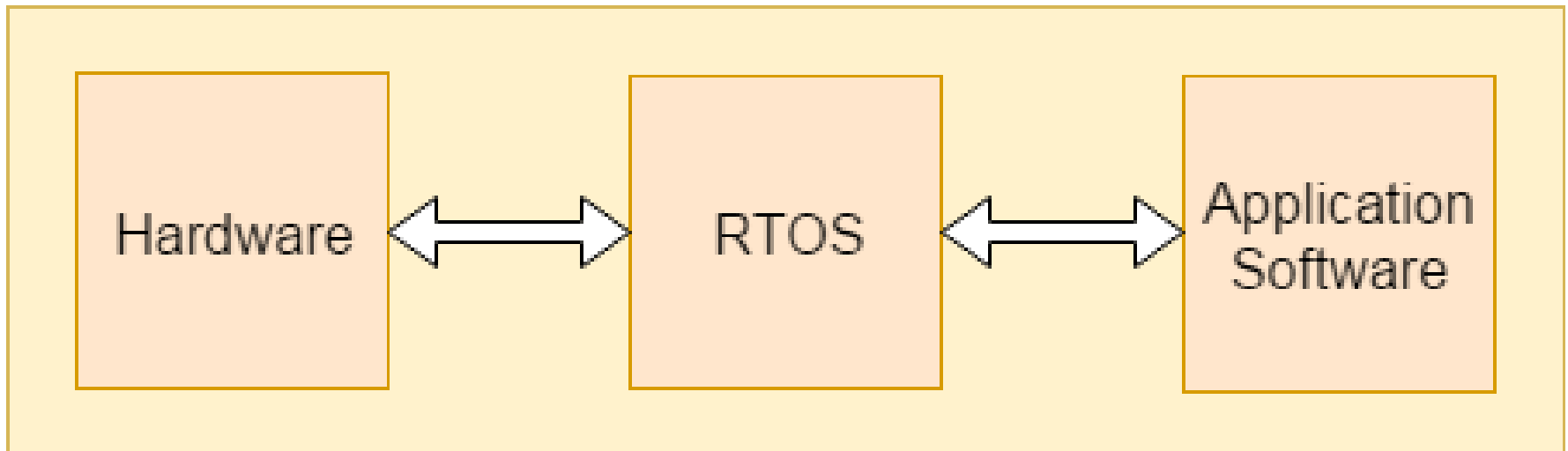
- Component that is physically connected with an embedded system.
- Comprises of Processors (microcontroller based integrated circuit), power supply, LCD display, Timer counter, memory, communication port

- **Application software:**

- Application software allows the user to perform varieties of application to be run on an embedded system by changing the code installed in an embedded system.

- **Real Time Operating system (RTOS):**
 - RTOS supervises the way an embedded system works.
 - It act as an interface between hardware and application software which supervises the application software and provide mechanism to let the processor run on the basis of scheduling for controlling the effect of latencies.
 - Examples of Embedded operating systems:
 - Linux based
(OpenWrt, Zeroshell, Android, LineageOS, LEDE, LibreCM C)
 - Others: NetBSD, ThreadX, FreeRTOS, QNX, VxWorks, INTEGRITY

Components of Embedded Control Systems



Characteristics of Embedded System

1. An embedded system is software embedded into computer hardware that makes a system dedicated to be used for variety of application.
2. Embedded system generally used for do specific task that provide real-time output on the basis of various characteristics of an embedded system.
3. Embedded system may contain a smaller part within a larger device that used for serving the more specific application to perform variety of task using hardware-software intermixing configuration.
4. It provides high reliability and real-time computation ability.

Categories of Embedded Systems

- **Hard real-time embedded systems**

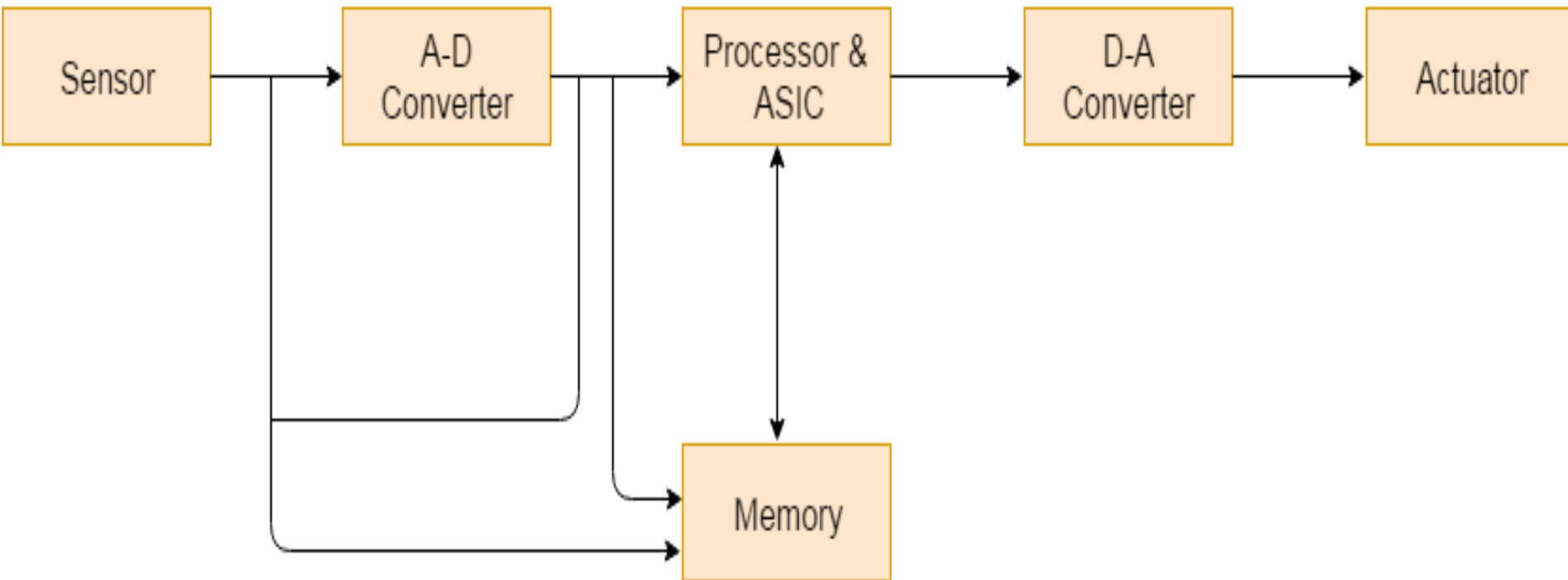
- They are designed with strict deadlines in their operations.
- The system is expected to complete task at a specified time. Failing to do so is considered system failure and the results could be catastrophic.
- Example is an air traffic control systems.

- **Soft real-time embedded systems**

- They are designed to reach deadlines but do not fail if a deadline is missed.
- They may degrade their quality of service in such an event to improve responsiveness.
- Example of soft real-time systems is multimedia streaming

Designing of an embedded system

- **Basic Structure of an Embedded System**



- **Sensor:** Sensor used for sensing the change in environment condition and it generate the electric signal on the basis of change in environment condition. Therefore it is also called as transducers for providing electric input signal on the basis of change in environment condition.
- **Analog-to-digital (A-D) converter** - a device that converts analog electric input signal into its equivalent digital signal for further processing
- **Processor & ASICs:** Processor used for processing the signal and data to execute desired set of instructions with high-speed of operation.
- **Application specific integrated circuit (ASIC)** is an integrated circuit designed to perform task specific operation inside an embedded system.

- **Digital-to-analog (D-A) Converter:** a device that converts digital electric input signal into its equivalent analog signal for further processing
- **Actuators:** a comparator used for comparing the analog input signal level to desired output signal level for providing the error free output from the system.

Microcontroller vrs a microprocessor

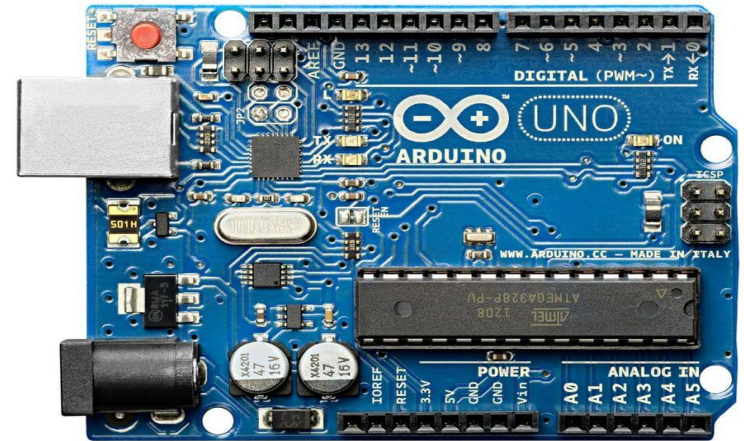
- A **microcontroller** is a compact integrated circuit **designed for specific tasks** in embedded systems, whereas a **microprocessor** is **a general-purpose computing device**.
- **Microcontrollers** typically include **a processor, memory, and I/O peripherals**, while **microprocessors** **require external components to perform similar tasks**.

Popular Microcontroller Platforms

- Arduino
- Raspberry Pi
- STM32
- PIC
- AVR.

Arduino

- The Arduino platform is a popular choice for microcontroller programming because of its **user-friendly interface and extensive community support.**
- **Advantages**
 - Its open-source nature
 - extensive library support
 - IDE environment and ease of use.
- **Disadvantages**
 - Arduino boards may not be suitable for all applications due to their limited processing power and memory storage.



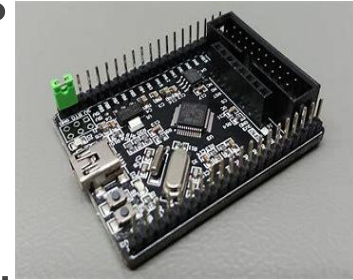
Raspberry Pi

- The Raspberry Pi platform is another popular choice for microcontroller programming, known for its versatility and powerful processing capabilities.
- Launched in 2012
- **Advantages**
 - its powerful processing capabilities and large community support.
 - Raspberry Pi has its own operating system known as “Raspbian” based on Linux distribution and installer for “Pi-apps’.
- **Disadvantages:**
 - However, Raspberry Pi boards may be more challenging for beginners due to their complexity, storage limitation and higher power consumption compared to other microcontroller platforms.



STM32

- The STM32 platform is a family of microcontrollers developed by STMicroelectronics
- It is known for its high-performance microcontrollers, which offer a range of features and capabilities suitable for various applications like printers and complex circuit boards
- The STM32 platform was first introduced in 2007
- Advantages
 - its high processing power, energy efficiency, and extensive peripheral support.
- Disadvantages:
 - However, the STM32 platform may have a steeper learning curve compared to Arduino and Raspberry Pi, making it more challenging for beginners.



PIC



- Developed by Microchip Technology
- First introduced in the early 1980s, PIC microcontrollers are known for their simplicity, and low cost, making them a popular choice for various applications in peripherals and audio systems.
- The advantages
 - Ease of use and extensive community support.
- Disadvantages:
 - Less suitable for complex applications due to limited processing power and memory compared to other platforms.

AVR



- The AVR microcontroller was developed by ATMEL in 1996 to support a wide range of hardware applications.
- **Advantages**
 - AVR microcontrollers include 8-16 bit timers, 8-bit and 16-bit timers, internal oscillators, 10-12 bit ADC and DAC
- **Disadvantages:**
 - AVR Microcontrollers offer low-power operation with a reduced clock speed ranging from 0 to 20 MHz.

Microcontroller Programming Languages

- C/C++
- Python
- Assembly-level languages.

